

JAZPET-RC

Jan S. Dam

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Series: series_name

This report documents the quality control analysis for the dataset using the NEMA/EARL phantom. The analysis includes geometric verification, Maximum Intensity Projections (MIPs) for alignment checks, intensity profile analysis, and quantitative recovery coefficient (RC) calculations.

Geometric Verification & MIPs

The following figures verify the geometric detection of spheres and their alignment in the scanner field of view. Original spheres are shown in solid red, and the 37 mm background VOIs (shifted -55 mm in Z) are shown in dashed blue in the Coronal view.

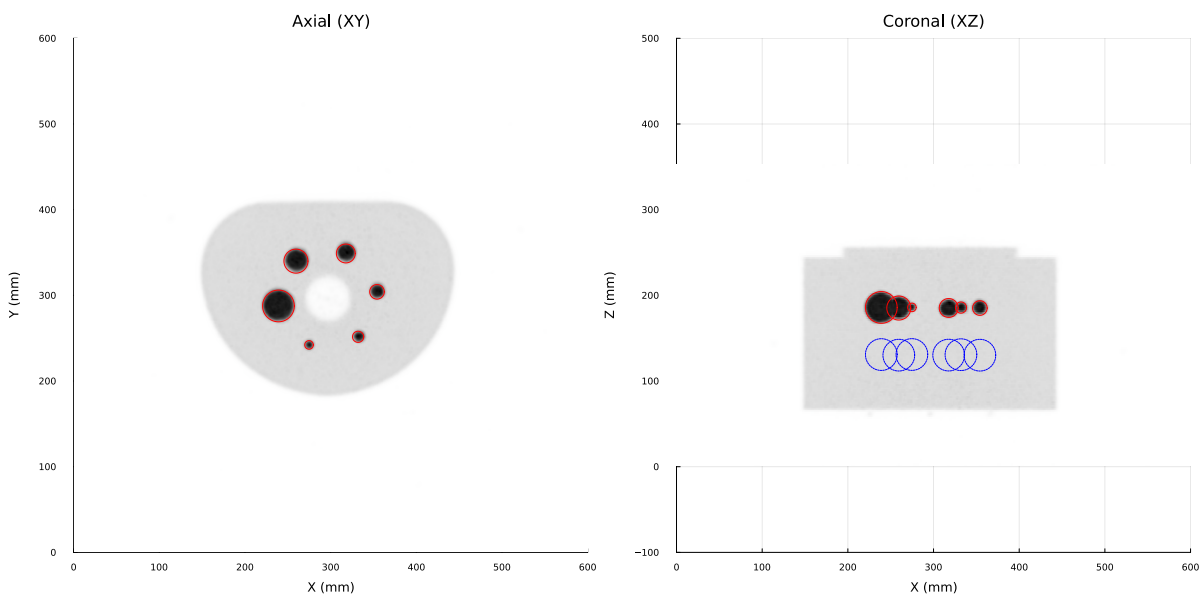


Figure 1: Axial (XY) and Coronal (XZ) Maximum Intensity Projections.

Manual Sphere Offsets

Diameter (mm)	Offset X (mm)	Offset Y (mm)	Offset Z (mm)
10.0	1.0	0.0	1.0
13.0	1.0	0.0	1.0
17.0	-1.0	0.0	2.0
22.0	0.0	0.0	-3.0
28.0	0.0	0.0	4.0
37.0	3.0	3.0	4.0

Sphere Intensity Profiles

Cross-sectional profiles through the center of selected spheres are shown below. The dashed line indicates the expected true activity concentration.

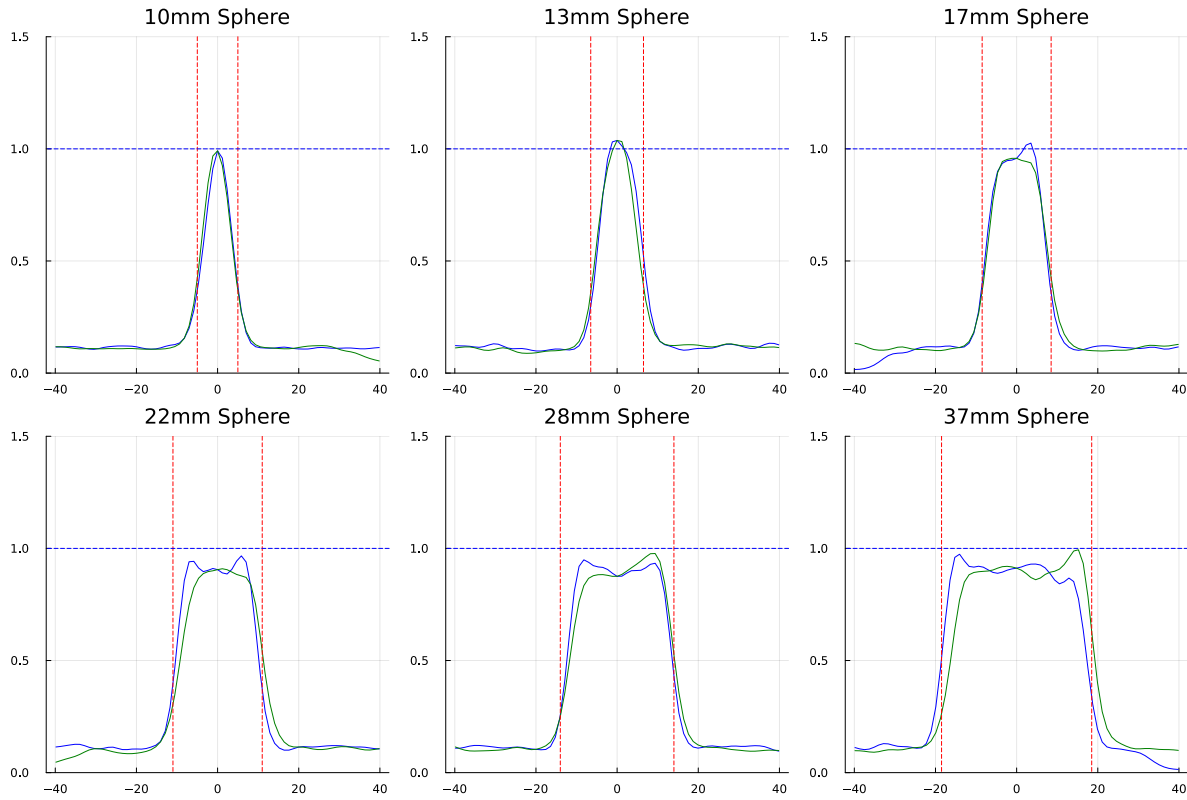


Figure 2: Intensity Profiles for Spheres (Relative RC). The x-axis represents the distance from the sphere center in millimeters (mm).

Recovery Coefficients

The table below lists the calculated Recovery Coefficients (RC) for all spheres. Background VOI (37 mm diameter, shifted -55 mm in Z) metrics are included.

Diameter (mm)	RC Phy	RC Mean	RC Max	RC Peak	RC BG
10.0	0.57	0.68	0.99	0.47	1.05
13.0	0.63	0.74	1.04	0.70	1.06
17.0	0.67	0.77	1.04	0.91	1.05
22.0	0.70	0.81	1.06	0.93	1.04
28.0	0.71	0.81	1.03	0.93	1.05
37.0	0.78	0.83	1.01	0.91	1.06

Recovery Curves

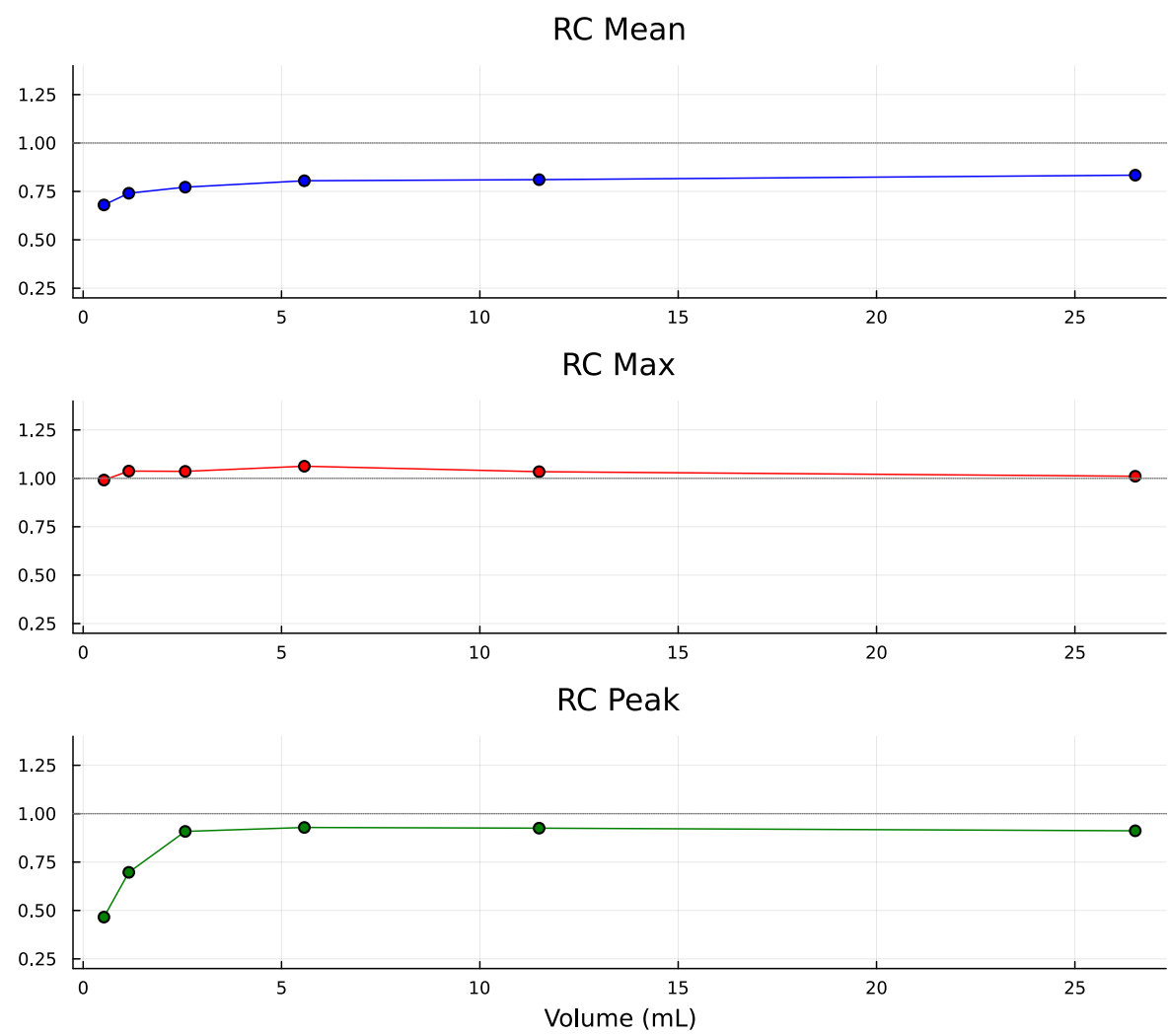


Figure 3: Recovery Coefficient curves

Calibration Factor and SNR

VOI Chamber	1	2	3	4	5	6	Total
Mean (kBq/mL)	2330.08	2351.32	2335.99	2312.39	2321.27	2359.61	2335.12
STD (kBq/mL)	168.93	183.77	173.84	179.78	180.10	175.77	177.84
SNR	13.79	12.79	13.44	12.86	12.89	13.42	13.13
RC BG	1.05	1.06	1.05	1.04	1.05	1.06	1.05